

TECHNICAL ARCHITECTURE DOCUMENTATION

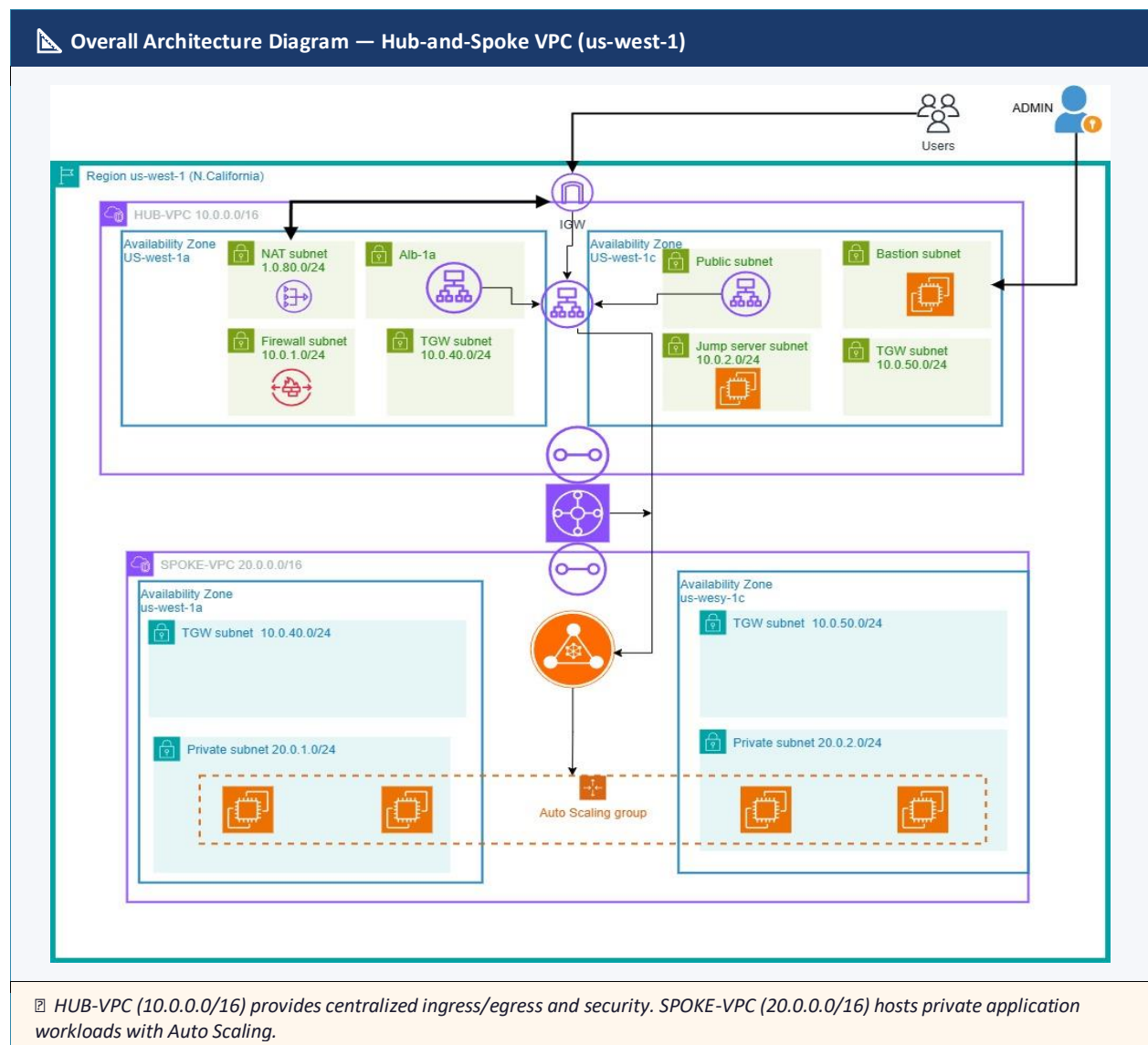
# AWS Hub-and-Spoke Architecture: Implementation Guide

Centralized Transit & Security Design |



# 1. Architecture Overview

The following diagram illustrates the complete Hub-and-Spoke network topology deployed in AWS Region us-west-1 (N. California). The architecture separates shared networking and security services (HUB-VPC) from application workloads (SPOKE-VPC), interconnected via an AWS Transit Gateway.



## 2. HUB-VPC — Centralized Transit & Security Layer

The HUB-VPC (vpc-05b0227dc3c6fad4d, CIDR: 10.0.0.0/16) is the central networking hub. It hosts all shared services including the Internet Gateway, NAT Gateway, AWS Network Firewall, ALB, Bastion Host, Jump Server, and Transit Gateway attachments across two Availability Zones.

### 2.1 HUB-VPC Resource Map

The Resource Map below shows the HUB-VPC subnet structure, route tables, and network connections as seen in the AWS Console. The HUB-VPC contains 4 subnets across two AZs with 5 route tables and 2 network connections (IGW and NAT).

#### HUB-VPC Resource Map — AWS Console (vpc-05b0227dc3c6fad4d)

**Details** Info

**VPC ID**  
vpc-05b0227dc3c6fad4d

**DNS resolution**  
Enabled

**Main network ACL**  
acl-0879ec48f176f0046

**IPv6 CIDR**  
—

**Encryption control ID**  
—

**State**  
Available

**Tenancy**  
default

**Default VPC**  
No

**Network Address Usage metrics**  
Disabled

**Encryption control mode**  
—

**Block Public Access**  
Off

**DHCP option set**  
dopt-066533031adc6b87d

**IPv4 CIDR**  
10.0.0.0/16

**Route 53 Resolver DNS Firewall rule groups**  
—

**DNS hostnames**  
Disabled

**Main route table**  
rtb-0800bac2b2c14a60e / firewall

**IPv6 pool**  
—

**Owner ID**  
897192298286

Resource map

CIDRs

Flow logs

Tags

Integrations

**Resource map** Info

Show all details

**VPC**  
Your AWS virtual network  
Hub-vpc

**Subnets (4)**  
Subnets within this VPC  
us-west-1a  
TGW-subnet  
Firewall  
us-west-1c  
Bastion  
private

**Route tables (5)**  
Route network traffic to resources  
firewall  
private  
NAT  
bastion  
tgw\_route

**Network Connections (2)**  
Connections to other networks  
Hub-igw  
NAT-HUB

Activate Windows  
Go to Settings to activate Windows

HUB-VPC has 4 subnets: TGW-subnet & Firewall (us-west-1a) · Bastion & private (us-west-1c). Route tables: firewall, private, NAT, bastion, tgw\_route.

### 2.2 HUB-VPC Subnet Inventory

| Availability Zone | Subnet Name     | CIDR / Role  | Purpose                          |
|-------------------|-----------------|--------------|----------------------------------|
| us-west-1a        | TGW-subnet      | 10.0.40.0/24 | Transit Gateway attachment point |
| us-west-1a        | Firewall        | 10.0.1.0/24  | AWS Network Firewall endpoint    |
| us-west-1a        | NAT Subnet      | 1.0.80.0/24  | NAT Gateway — EIP 54.153.12.117  |
| us-west-1a        | ALB-1a (Public) | —            | Application Load Balancer        |
| us-west-1c        | Jump            | 10.0.2.0/24  | Jump Server for internal access  |

| Availability Zone | Subnet Name   | CIDR / Role  | Purpose                                   |
|-------------------|---------------|--------------|-------------------------------------------|
| us-west-1c        | Public Subnet | —            | Public-facing resources                   |
| us-west-1c        | TGW Subnet    | 10.0.50.0/24 | Transit Gateway attachment (secondary AZ) |

## 2.3 . Route Table Details

### 2.3.1 Firewall Route Table

| Routes | Subnet associations | Edge associations | Route propagation | Tags |
|--------|---------------------|-------------------|-------------------|------|
|--------|---------------------|-------------------|-------------------|------|

| Routes (3)                                                                            |                                       |        |            |                    |  |  |
|---------------------------------------------------------------------------------------|---------------------------------------|--------|------------|--------------------|--|--|
| <input type="text" value="Filter routes"/> <span>Both</span> <span>Edit routes</span> |                                       |        |            |                    |  |  |
| Destination                                                                           | Target                                | Status | Propagated | Route Origin       |  |  |
| 0.0.0.0/0                                                                             | <a href="#">nat-1dfeb7a5dd60d2707</a> | Active | No         | Create Route       |  |  |
| 10.0.0.0/16                                                                           | local                                 | Active | No         | Create Route Table |  |  |
| 20.0.0.0/16                                                                           | <a href="#">tgw-0b02dde0fe83b887b</a> | Active | No         | Create Route       |  |  |

The Firewall RT controls traffic that has been inspected by the Network Firewall. After inspection: internet-bound traffic (0.0.0.0/0) is forwarded to the NAT Gateway, local Hub traffic (10.0.0.0/16) stays local, and Spoke-bound traffic (20.0.0.0/0) is sent back through the TGW.

### 2.3.2 NAT Route Table

| Routes (3)                                                                            |                                        |        |            |              |  |  |
|---------------------------------------------------------------------------------------|----------------------------------------|--------|------------|--------------|--|--|
| <input type="text" value="Filter routes"/> <span>Both</span> <span>Edit routes</span> |                                        |        |            |              |  |  |
| Destination                                                                           | Target                                 | Status | Propagated | Route Origin |  |  |
| 0.0.0.0/0                                                                             | <a href="#">igw-03ab192f3c871d941</a>  | Active | No         | Create Route |  |  |
| 10.0.0.0/16                                                                           | <a href="#">vpce-06b39077c6ee777d3</a> | Active | No         | Create Route |  |  |
| 20.0.0.0/16                                                                           | <a href="#">tgw-0b02dde0fe83b887b</a>  | Active | No         | Create Route |  |  |

The NAT RT routes internet-bound traffic (0.0.0.0/0) through the Internet Gateway (IGW). It also routes Hub-local (10.0.0.0/16) via the VPC Firewall endpoint and Spoke-bound (20.0.0.0/16) back through the TGW — ensuring return traffic from the internet flows correctly back to private Spoke instances.

### 2.3.3 Bastion Route Table

| Routes (3)                                                                            |                                       |        |            |                    |  |  |
|---------------------------------------------------------------------------------------|---------------------------------------|--------|------------|--------------------|--|--|
| <input type="text" value="Filter routes"/> <span>Both</span> <span>Edit routes</span> |                                       |        |            |                    |  |  |
| Destination                                                                           | Target                                | Status | Propagated | Route Origin       |  |  |
| 0.0.0.0/0                                                                             | <a href="#">igw-03ab192f3c871d941</a> | Active | No         | Create Route       |  |  |
| 10.0.0.0/16                                                                           | local                                 | Active | No         | Create Route Table |  |  |
| 20.0.0.0/16                                                                           | <a href="#">tgw-0b02dde0fe83b887b</a> | Active | No         | Create Route       |  |  |

The Bastion RT gives the Bastion Host subnet direct internet access (0.0.0.0/0 → IGW) so admins can SSH into it from outside. It also routes Hub-internal (10.0.0.0/16) locally and sends Spoke-bound traffic (20.0.0.0/16) through the TGW, enabling the Bastion to SSH-hop into private Spoke instances.

2.3.4 TGW Route Table

Routes (3)

Filter routes

< 1 >

| Destination | Target                                 | Status | Propagated | Route Origin       |
|-------------|----------------------------------------|--------|------------|--------------------|
| 0.0.0.0/0   | <a href="#">vpce-06b39077c6ee777d3</a> | Active | No         | Create Route       |
| 10.0.0.0/16 | local                                  | Active | No         | Create Route Table |
| 20.0.0.0/16 | <a href="#">tgw-0b02dde0fe83b887b</a>  | Active | No         | Create Route       |

The TGW RT handles traffic arriving from the Transit Gateway. Internet-bound traffic (0.0.0.0/0) is sent to the Firewall VPC endpoint for inspection before continuing. Hub-local traffic stays local. Spoke-bound traffic (20.0.0.0/16) is forwarded back through the TGW to reach Spoke instances

3. SPOKE-VPC

The SPOKE-VPC (vpce-0b7a6cdfb95557591, CIDR: 20.0.0.0/16) hosts all application workloads in private subnets. It connects to the HUB-VPC exclusively through the Transit Gateway. The main route table is spoke-RT.

3.1 SPOKE-VPC Resource Map

SPOKE-VPC Resource Map — AWS Console (vpce-0b7a6cdfb95557591)

vpce-0b7a6cdfb95557591 / Spoke-Vpc

Actions

Details

VPC ID

vpce-0b7a6cdfb95557591

DNS resolution

Enabled

Main network ACL

acl-0c0e496f0bdf77262

IPv6 CIDR

-

Encryption control ID

-

State

Available

Tenancy

default

Default VPC

No

Network Address Usage metrics

Disabled

Encryption control mode

-

Block Public Access

Off

DHCP option set

dopt-066533031adc6b87d

IPv4 CIDR

20.0.0.0/16

Route 53 Resolver DNS Firewall rule groups

-

DNS hostnames

Disabled

Main route table

rtb-0c3c23a15a8fbd140 / spoke-RT

IPv6 pool

-

Owner ID

897192298286

Resource map

CIDRs

Flow logs

Tags

Integrations

Resource map

Subnets (3)

Route tables (2)

Network Connections (1)

VPC

Your AWS virtual network

Spoke-Vpc

Subnets within this VPC

us-west-1a

public-subnet

private-subnet1

us-west-1c

private-subnet2

Route network traffic to resources

test

spoke-RT

Connections to other networks

test

Activate Windows

Go to Settings to activate Windows

SPOKE-VPC has 3 subnets: public-subnet & private-subnet1 (us-west-1a) · private-subnet2 (us-west-1c). Route tables: test, spoke-RT. Connection: TGW.

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## Route Table: Spoke RT

| Routes                                                                                      | Subnet associations                   | Edge associations | Route propagation | Tags               |
|---------------------------------------------------------------------------------------------|---------------------------------------|-------------------|-------------------|--------------------|
| <div>Routes (3)<div>BothEdit routes</div><div>Filter routes</div><div>&lt;1&gt;</div></div> |                                       |                   |                   |                    |
| Destination                                                                                 | Target                                | Status            | Propagated        | Route Origin       |
| 0.0.0.0/0                                                                                   | <a href="#">tgw-0b02dde0fe83b887b</a> | Active            | No                | Create Route       |
| 10.0.0.0/16                                                                                 | <a href="#">tgw-0b02dde0fe83b887b</a> | Active            | No                | Create Route       |
| 20.0.0.0/16                                                                                 | local                                 | Active            | No                | Create Route Table |

The Spoke RT sends all traffic — both internet-bound (0.0.0.0/0) and Hub-bound (10.0.0.0/16) — through the Transit Gateway. The Spoke VPC has no local internet path, so the TGW is the only exit. The 20.0.0.0/16 local route keeps intra-Spoke traffic within the VPC without going through the TGW.

### 3.2 SPOKE-VPC Subnet Inventory

| Availability Zone | Subnet Name      | CIDR         | Purpose                                   |
|-------------------|------------------|--------------|-------------------------------------------|
| us-west-1a        | TGW Subnet       | 10.0.40.0/24 | Transit Gateway attachment point          |
| us-west-1a        | Private Subnet 1 | 20.0.1.0/24  | Application EC2 instances — AZ-a          |
| us-west-1c        | TGW Subnet       | 10.0.50.0/24 | Transit Gateway attachment (secondary AZ) |
| us-west-1c        | Private Subnet 2 | 20.0.2.0/24  | Application EC2 instances — AZ-c          |

## 4. Transit Gateway — Centralized Routing

The Transit Gateway (tgw-0b02dde0fe83b887b / Transit-gateway-team2) is the routing backbone of the architecture. It connects the HUB-VPC and SPOKE-VPC with full route propagation and a static default route (0.0.0.0/0) pointing to the HUB-VPC for centralized egress.

## 4.1 Transit Gateway Details

**Transit Gateway Details — tgw-0b02dde0fe83b887b**

**tgw-0b02dde0fe83b887b / Transit-gateway-team2** [Info](#) [Actions](#)

| Details                                                                                                                |                                                  |                                                                                |                                                  |
|------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------|
| <b>Transit gateway ID</b><br><a href="#">tgw-0b02dde0fe83b887b</a>                                                     | <b>State</b><br>Available                        | <b>Amazon ASN</b><br><a href="#">64512</a>                                     | <b>DNS support</b><br>Enable                     |
| <b>Transit gateway ARN</b><br><a href="#">arn:aws:ec2:us-west-1:897192298286:transit-gateway/tgw-0b02dde0fe83b887b</a> | <b>Default association route table</b><br>Enable | <b>Association route table ID</b><br><a href="#">tgw-rtb-0669ff68ed77c65d7</a> | <b>Auto accept shared attachments</b><br>Disable |
| <b>Owner ID</b><br><a href="#">897192298286</a>                                                                        | <b>Default propagation route table</b><br>Enable | <b>Propagation route table ID</b><br><a href="#">tgw-rtb-0669ff68ed77c65d7</a> | <b>VPN ECMP support</b><br>Enable                |
| <b>Description</b><br>team2                                                                                            | <b>Transit gateway CIDR blocks</b><br>-          | <b>Security Group Referencing support</b><br>Disable                           | <b>Metering policy ID</b><br>-                   |
| <b>Multicast support</b><br>Disable                                                                                    | <b>Encryption support</b><br>Disabled            | <b>Encryption state message</b><br>-                                           |                                                  |

**TGW ID:** [tgw-0b02dde0fe83b887b](#) · **Amazon ASN:** [64512](#) · **State:** Available · **DNS support:** Enabled · **Description:** team2

## 4.2 TGW Attachments

**Transit Gateway Attachments — Hub & Spoke**

|       | Attachment ID                                | Transit Gateway ID                    | State     | Resource Type | VPC ID                                |
|-------|----------------------------------------------|---------------------------------------|-----------|---------------|---------------------------------------|
| Hub   | <a href="#">tgw-attach-00025d4fb53c13d25</a> | <a href="#">tgw-0b02dde0fe83b887b</a> | Available | VPC           | <a href="#">vpc-05b0227dc3c6fad4d</a> |
| SPOKE | <a href="#">tgw-attach-0e4e6b53aaf6f0851</a> | <a href="#">tgw-0b02dde0fe83b887b</a> | Available | VPC           | <a href="#">vpc-0b7a6cdfb95557591</a> |

**Two attachments:** Hub ([tgw-attach-00025d4fb53c13d25](#) → [vpc-05b0227dc3c6fad4d](#)) and SPOKE ([tgw-attach-0e4e6b53aaf6f0851](#) → [vpc-0b7a6cdfb95557591](#)) — both Active.

## 4.3 TGW Route Table

**Transit Gateway Route Table — 3 Active Routes**

**Routes (3)** [Info](#) [Actions](#) [Create static route](#)

Find route by attribute or tag

| CIDR        | Attachment ID                                | Resource ID                           | Resource type | Route type | Route state | Prefix list ID |
|-------------|----------------------------------------------|---------------------------------------|---------------|------------|-------------|----------------|
| 0.0.0.0/0   | <a href="#">tgw-attach-00025d4fb53c13d25</a> | <a href="#">vpc-05b0227dc3c6fad4d</a> | VPC           | Static     | Active      | -              |
| 10.0.0.0/16 | <a href="#">tgw-attach-00025d4fb53c13d25</a> | <a href="#">vpc-05b0227dc3c6fad4d</a> | VPC           | Propagated | Active      | -              |
| 20.0.0.0/16 | <a href="#">tgw-attach-0e4e6b53aaf6f0851</a> | <a href="#">vpc-0b7a6cdfb95557591</a> | VPC           | Propagated | Active      | -              |

Activate Windows

**Routes:** 0.0.0.0/0 (Static → HUB) · 10.0.0.0/16 (Propagated → HUB) · 20.0.0.0/16 (Propagated → SPOKE) — all Active.

## 5. NAT Gateway

The NAT Gateway (nat-1dfb7a5dd60d2707 / NAT-HUB) provides outbound internet access for all private resources in both HUB-VPC and SPOKE-VPC. It is deployed in us-west-1a with a public Elastic IP address (54.153.12.117).

**NAT Gateway Details — nat-1dfeb7a5dd60d2707 / NAT-HUB**

### nat-1dfeb7a5dd60d2707 / NAT-HUB Actions ▾

#### Details

**NAT gateway ID**  
 nat-1dfeb7a5dd60d2707

**NAT gateway ARN**  
 arn:aws:ec2:us-west-1:897192298286:natgateway/nat-1dfeb7a5dd60d2707

**VPC**  
 vpc-05b0227dc3c6fad4d / Hub-vpc

**Availability mode**  
Regional

**Connectivity type**  
Public

**Method of EIP allocation**  
Manual

**State**  
✔ Available

**Created**  
 Tuesday, June 16, 2026 at 23:45:27 GMT+3

**State message** [Info](#)  
—

**Deleted**  
—

[IP addresses](#) | [Monitoring](#) | [Flow logs](#) | [Tags](#)

#### Associated IP addresses (1) Edit IP address associations

| IP address ▾  | Status ▾    | Availability Zone ▾   | Allocation ID ▾            | Association ID ▾           |
|---------------|-------------|-----------------------|----------------------------|----------------------------|
| 54.153.12.117 | ✔ Succeeded | usw1-az1 (us-west-1a) | eipalloc-0eb57a833f07eed4d | eipassoc-027bd97bc88e39136 |

**NAT-HUB · Connectivity:** Public · **State:** Available · **AZ:** usw1-az1 (us-west-1a) · **EIP:** 54.153.12.117 · **VPC:** Hub-vpc

## 6. AWS Network Firewall

The AWS Network Firewall is deployed in the Firewall Subnet (us-west-1a) of the HUB-VPC. It performs stateful traffic inspection on all traffic passing through the hub. The associated policy is 'policy2' which is in sync with the firewall endpoint.



## 6.1 Firewall Endpoints & Policy

**Network Firewall — Endpoints & Policy**

**Firewall endpoints**

| Availability Zone | Firewall subnet          | Endpoint ID            | Firewall endpoint status |
|-------------------|--------------------------|------------------------|--------------------------|
| us-west-1a        | subnet-09d41c871fb25de36 | vpce-06b39077c6ee777d3 | Ready                    |

**Firewall Policy** [Edit](#)

Associated firewall policy  
[policy2](#)

**Configuration sync state**

Find by keyword

| Resource name | Resource type   | Availability Zone | Sync state |
|---------------|-----------------|-------------------|------------|
| policy2       | Firewall policy | us-west-1a        | In sync    |

**Endpoint:** vpce-06b39077c6ee777d3 · **Subnet:** subnet-09d41c871fb25de36 (us-west-1a) · **Status:** Ready · **Policy:** policy2 · **Sync state:** In sync

## 6.2 Firewall Policy (policy2)

**Firewall Policy — policy2 Rule Groups**

[policies](#) > [policy2](#)

| Priority                                                           | Name | Capacity |
|--------------------------------------------------------------------|------|----------|
| No stateless rule groups                                           |      |          |
| Choose Add rule groups to add stateless rule groups to the policy. |      |          |

**Stateful rule evaluation order and default actions** [Edit](#)

The way that your stateful rules are ordered for evaluation.

Rule order

Action order

Default actions

**Stateful rule groups (0)** [Actions](#)

| Name                                                              | Capacity | Managed | Run in alert mode |
|-------------------------------------------------------------------|----------|---------|-------------------|
| No stateful rule groups                                           |          |         |                   |
| Choose Add rule groups to add stateful rule groups to the policy. |          |         |                   |

**Capacity units consumed by stateless rule groups**  
The total capacity units consumed by stateless rule groups can't exceed 30,000.  
0/30,000

**Capacity units consumed by stateful rule groups**  
The total capacity units consumed by stateful rule groups can't exceed 30,000.  
0/30,000

Activate Windows  
Go to Settings to activate Windows.

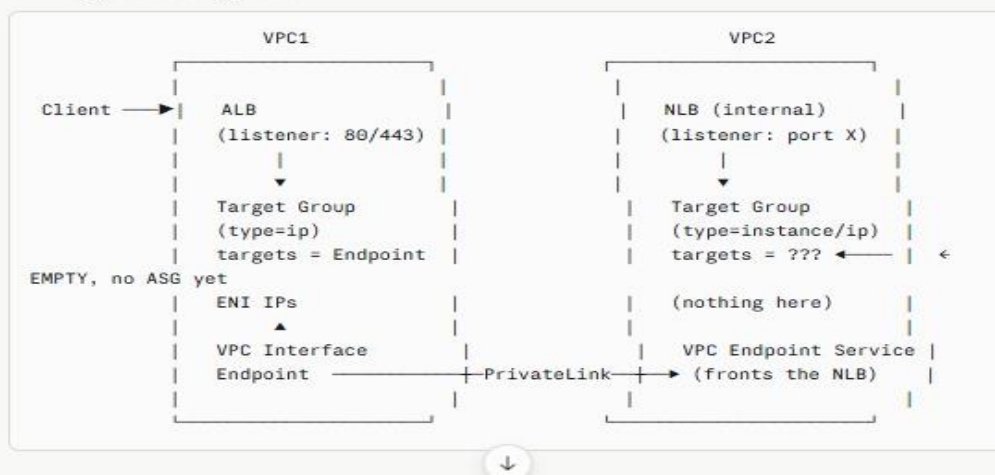
Policy2 uses Action Order for stateful rule evaluation. Currently 0 stateless and 0 stateful rule groups configured — in open/pass-through mode for testing. Rule groups can be added to enforce domain filtering and IP blocking.

## 7. Load Balancers — ALB & NLB

The architecture implements a two-tier load balancing strategy: an Application Load Balancer (ALB) in the HUB-VPC handles external HTTP/HTTPS traffic from users, while a Network Load Balancer (NLB) in the SPOKE-VPC distributes traffic to application EC2 instances

### 7.1 Architecture: ALB → Private Link → NLB Flow

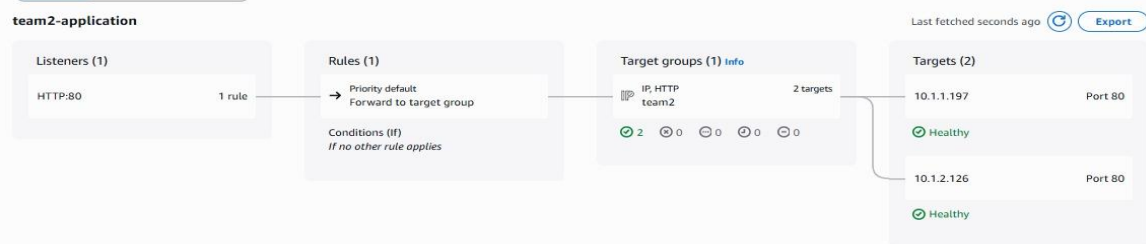
#### Load Balancer Architecture — ALB to NLB via PrivateLink



Client → ALB (VPC1, port 80/443) → Target Group (type=ip, ENIs) → VPC Interface Endpoint → PrivateLink → VPC Endpoint Service → NLB (VPC2, internal)

### 7.2 Application Load Balancer (ALB) — team2-application

#### ALB Detail — team2-application



Listener: HTTP:80 · Rule: Forward to target group 'team2' (type=IP) · 2 healthy targets

## ✓ ALB Target Group — 2 Healthy Targets

**Registered targets (2)** [Info](#) Anomaly mitigation: **Not applicable** [Deregister](#) [Register targets](#)

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

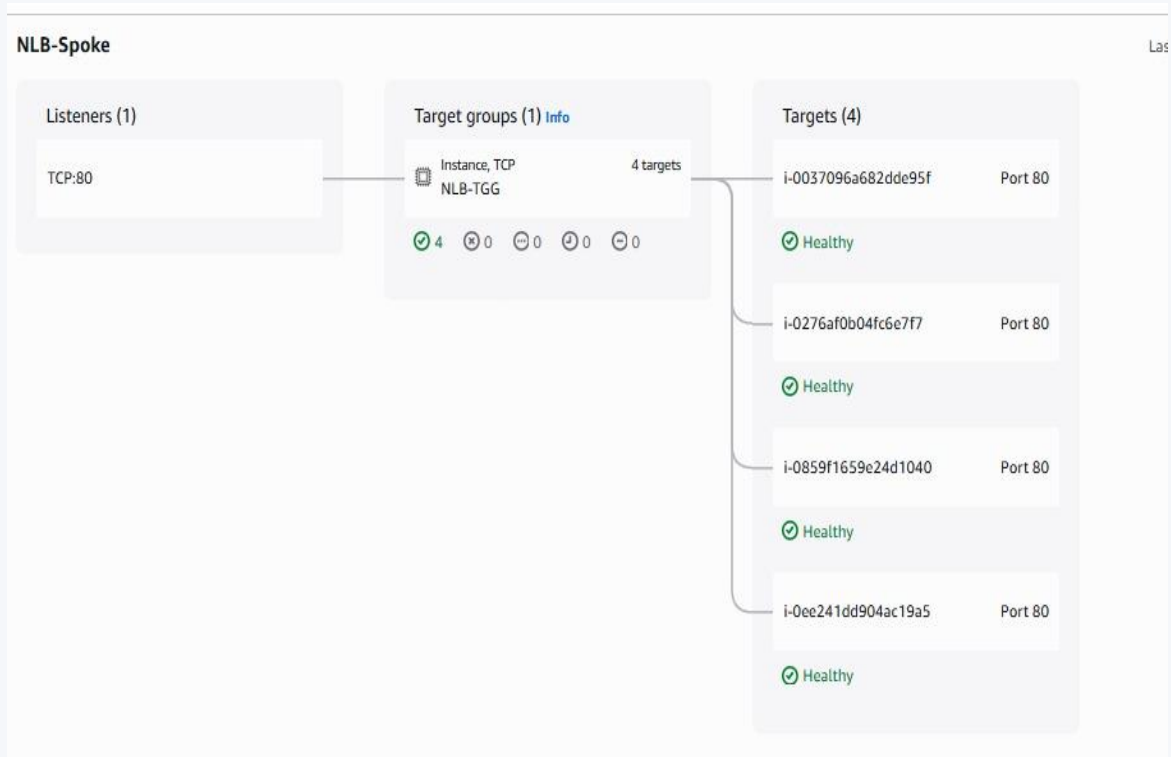
Filter targets

| <input type="checkbox"/> | IP address | Port | Zone | Health status | Health status details | Administrative override | Override details                      | Anomaly detection... |
|--------------------------|------------|------|------|---------------|-----------------------|-------------------------|---------------------------------------|----------------------|
| <input type="checkbox"/> | 10.1.1.197 | 80   | All  | Healthy       | -                     | No override             | No override is currently active on... | Normal               |
| <input type="checkbox"/> | 10.1.2.126 | 80   | All  | Healthy       | -                     | No override             | No override is currently active on... | Normal               |

🔗 Target IPs: 10.1.1.197:80 (Healthy) · 10.1.2.126:80 (Healthy) — these are the ENI IPs of the VPC Interface Endpoint

## 7.3 Network Load Balancer (NLB) — NLB-Spoke

### 🏠 NLB-Spoke — Listener, Target Group & Targets



🔗 NLB-Spoke · Listener: TCP:80 · Target Group: NLB-TGG (type=Instance) · 4 Healthy targets across both AZs

## ✓ NLB Target Group — 4 Healthy Instances

Targets

Monitoring

Health checks

Attributes

Tags

Registered targets (4)

Q

Filter targets

<

1

>

| <input type="checkbox"/> | Instance ID                        | Name | Port | Zone                | Health status | Health status de... | Administrativ... | Override details      | Launch time         |
|--------------------------|------------------------------------|------|------|---------------------|---------------|---------------------|------------------|-----------------------|---------------------|
| <input type="checkbox"/> | <a href="#">i-0ee241dd904a...</a>  | -    | 80   | us-west-1c (usw1... | Healthy       |                     | No override      | No override is cur... | June 21, 2026, 2... |
| <input type="checkbox"/> | <a href="#">i-0276af0b04fc6...</a> | -    | 80   | us-west-1a (usw...  | Healthy       |                     | No override      | No override is cur... | June 21, 2026, 2... |
| <input type="checkbox"/> | <a href="#">i-0859f1659e24...</a>  | -    | 80   | us-west-1a (usw...  | Healthy       |                     | No override      | No override is cur... | June 21, 2026, 2... |
| <input type="checkbox"/> | <a href="#">i-0037096a682d...</a>  | -    | 80   | us-west-1c (usw1... | Healthy       |                     | No override      | No override is cur... | June 21, 2026, 2... |

✎ 4 EC2 instances registered: i-0ee241dd904a, i-0276af0b04fc, i-0859f1659e24, i-0037096a682d — all Healthy on port 80 across us-west-1a and us-west-1c

## 8. Connectivity Validation

The following terminal outputs demonstrate successful end-to-end connectivity validation from EC2 instances in the SPOKE-VPCE. Tests confirm: cross-VPCE routing via TGW, outbound internet access via NAT Gateway, and successful cross-AZ communication.

### 8.1 Cross-VPCE Ping Test (SPOKE → HUB)

#### Terminal: Cross-VPCE Ping — ec2-user@ip-20-0-1-179

```

Jun 17, 2:46 AM
ec2-user@ip-20-0-1-179:~$ ping 10.0.2.180
PING 10.0.2.180 (10.0.2.180) 56(84) bytes of data:
64 bytes from 10.0.2.180: icmp_seq=1 ttl=126 time=1.48 ms
64 bytes from 10.0.2.180: icmp_seq=2 ttl=126 time=1.24 ms
64 bytes from 10.0.2.180: icmp_seq=3 ttl=126 time=1.84 ms
64 bytes from 10.0.2.180: icmp_seq=4 ttl=126 time=1.95 ms
64 bytes from 10.0.2.180: icmp_seq=5 ttl=126 time=2.26 ms
64 bytes from 10.0.2.180: icmp_seq=6 ttl=126 time=1.76 ms
64 bytes from 10.0.2.180: icmp_seq=7 ttl=126 time=1.66 ms
64 bytes from 10.0.2.180: icmp_seq=8 ttl=126 time=1.59 ms
64 bytes from 10.0.2.180: icmp_seq=9 ttl=126 time=1.22 ms
64 bytes from 10.0.2.180: icmp_seq=10 ttl=126 time=1.69 ms
64 bytes from 10.0.2.180: icmp_seq=11 ttl=126 time=2.11 ms
64 bytes from 10.0.2.180: icmp_seq=12 ttl=126 time=1.51 ms
64 bytes from 10.0.2.180: icmp_seq=13 ttl=126 time=2.25 ms
64 bytes from 10.0.2.180: icmp_seq=14 ttl=126 time=1.91 ms
^C
--- 10.0.2.180 ping statistics ---
14 packets transmitted, 14 received, 0% packet loss, time 13022ms
rtt min/avg/max/mdev = 1.216/1.747/2.261/0.317 ms
[ec2-user@ip-20-0-1-179 ~]$ ping 10.0.3.90
PING 10.0.3.90 (10.0.3.90) 56(84) bytes of data:
64 bytes from 10.0.3.90: icmp_seq=1 ttl=126 time=1.82 ms
64 bytes from 10.0.3.90: icmp_seq=2 ttl=126 time=1.29 ms
64 bytes from 10.0.3.90: icmp_seq=3 ttl=126 time=1.74 ms
64 bytes from 10.0.3.90: icmp_seq=4 ttl=126 time=1.30 ms
64 bytes from 10.0.3.90: icmp_seq=5 ttl=126 time=1.48 ms
^C
--- 10.0.3.90 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 1.285/1.525/1.819/0.220 ms
[ec2-user@ip-20-0-1-179 ~]$

```

🔗 From SPOKE EC2 (20.0.1.179): ping 10.0.2.180 (HUB Jump Server) — 14/14 packets received, 0% loss, avg RTT 1.747ms. ping 10.0.3.90 — 5/5 packets received, 0% loss. TGW routing confirmed working.

## 8.2 Outbound Internet Access via NAT Gateway

### Terminal: curl & ping google.com — Outbound NAT Confirmed

```

Jun 17, 2:47 AM
ec2-user@ip-20-0-1-179:~$ curl google.com
<HTML><HEAD><meta http-equiv="content-type" content="text/html; charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="http://www.google.com/">here</A>.
</BODY></HTML>
ec2-user@ip-20-0-1-179:~$ ping google.com
PING google.com (142.251.218.110) 56(84) bytes of data:
64 bytes from pnsfoa-ab-in-f14.1e100.net (142.251.218.110): icmp_seq=1 ttl=116 time=2.48 ms
64 bytes from pnsfoa-ab-in-f14.1e100.net (142.251.218.110): icmp_seq=2 ttl=116 time=1.75 ms
64 bytes from pnsfoa-ab-in-f14.1e100.net (142.251.218.110): icmp_seq=3 ttl=116 time=1.61 ms
64 bytes from pnsfoa-ab-in-f14.1e100.net (142.251.218.110): icmp_seq=4 ttl=116 time=1.55 ms
64 bytes from pnsfoa-ab-in-f14.1e100.net (142.251.218.110): icmp_seq=5 ttl=116 time=1.15 ms
^C
--- google.com ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.153/1.709/2.484/0.434 ms
ec2-user@ip-20-0-1-179:~$
  
```

🔗 From SPOKE EC2 (20.0.1.179): curl google.com → HTTP 301 response received. ping google.com (142.251.218.110) → 5/5 packets, 0% loss, avg RTT 1.709ms. NAT Gateway routing confirmed.

## 8.3 Connectivity Test Summary

| Test             | Source                | Destination             | Result            | RTT         |
|------------------|-----------------------|-------------------------|-------------------|-------------|
| Cross-VPC ping   | SPOKE 20.0.1.179      | HUB 10.0.2.180          | ✓ PASS — 0% loss  | avg 1.747ms |
| Internet curl    | SPOKE 20.0.1.179      | google.com              | ✓ PASS — HTTP 301 | —           |
| ALB Health Check | ALB team2-application | 10.1.1.197 & 10.1.2.126 | ✓ Healthy         | —           |
| NLB Health Check | NLB-Spoke             | 4 EC2 instances         | ✓ All Healthy     | —           |

## 9. Security Architecture Summary

| Layer                | Component               | Status      | Details                                    |
|----------------------|-------------------------|-------------|--------------------------------------------|
| Perimeter            | Internet Gateway        | ✓ Active    | Single ingress/egress for HUB-VPC          |
| Network Inspection   | AWS Network Firewall    | ✓ Ready     | vpce-06b39077c6ee777d3 · policy2 · In sync |
| Outbound NAT         | NAT Gateway             | ✓ Available | EIP 54.153.12.117 · Regional mode          |
| Admin Access         | Bastion Host            | ✓ Deployed  | us-west-1c · hardened admin entry          |
| Internal Hop         | Jump Server             | ✓ Deployed  | 10.0.2.x — cross-VPC SSH/RDP               |
| Routing Isolation    | Transit Gateway         | ✓ Available | tgw-0b02dde0fe83b887b · 3 active routes    |
| Traffic Distribution | ALB (team2-application) | ✓ Healthy   | HTTP:80 · 2 healthy IP targets             |
| Internal LB          | NLB-Spoke               | ✓ Healthy   | TCP:80 · 4 healthy instance targets        |

**Document prepared by Mostafa Adel**  
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